Lecture 14

Corey Powell

2026-10-26

Lecture 14

1. Independent Random Variables
2. Standard Error
3. Covariance
4. Reading Summary – Chapter 16
5. The Law of Averages

Read Chapter 18 for next time.

Quiz 7 is next Wednesday, November 4, in lecture.

Independent Random Variables

- Two random variables X and Y are **independent** if $P(X = x|Y = y) = P(X = x)$ for any values of x and y.
  - The values of Y don't have any influence on the values of X, and vice versa.
- This implies that

$$P(X = x \text{ and } Y = y) = P(X = x)P(Y = y)$$

  for any values of x and y.
- It follows that $E(XY) = E(X) * E(Y)$.

Standard Error

- Let X be a random variable.
- The **variance** of X is $\mathrm{Var}(X) = E\left[(X - E(X))^2\right]$.
  - The **standard error** of X is $SE(X) = \sqrt{\mathrm{Var}(X)}$.
- These formulas are analogous to the formulas for the variance and standard deviation of a list of numbers.

Standard Error Example

- Suppose a box has tickets $\{0, 4, 4, 8\}$.
- You draw a ticket and call it X.
- The expected value of X is 4, so $\mathrm{Var}(X) = E\left[(X-4)^2\right]$
- The possible values of X are $0, 4, 8$, so the possible values of $(X-4)^2$ are $0, 16$.
- $P\left[(X-4)^2 = 0\right] = 1/2$ and $P\left[(X-4)^2 = 16\right] = 1/2$
- $E\left[(X-4)^2\right] = 0 \times (1/2) + 16 \times (1/2) = 8$
- Thus $\mathrm{Var}(X) = 8$ and $SE(X) = 2\sqrt{2}$.

Standard Error Interpretation

- $SE(X)$ can be thought of as the long run standard deviation of X.
- The standard error tells us by about how much we might expect X to differ from its expected value.
- The standard error is a measure of the size of the chance variability in X.
- The variance is theoretically useful, but is **not** a measure of the size of chance variability.
- The standard error is just an ordinary number. It is *not* random.
- If we were to think of a list of numbers as a random variable Y as before, then what would be $SE(Y)$?

Standard Error Interpretation Using R

```
#Simulate 100,000,000 draws for random variable
#from 2 slides ago
p = runif(100000000)
X = 0*(p<1/4) + 4*(p>=1/4 & p<3/4) + 8*(p>=3/4)
#SE(X)
print(sd(X)*sqrt(99999999/100000000))
```

```
## [1] 2.828268
```

```
#Should be approximately 2*sqrt(2)
print(2*sqrt(2))
```

```
## [1] 2.828427
```

Covariance

- Let X and Y be a random variables.
- The **covariance** of X and Y is

$$\mathrm{Cov}(X, Y) = E\left[(X - E(X))(Y - E(Y))\right].$$

- The **correlation** of X and Y is $\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X,Y)}{SE(X)SE(Y)}$.
- These formulas are analogous to the formulas for covariance and correlation with a list of numbers.

Covariance Example 1

- Suppose a box has tickets $0, 4, 4, 8$.
- You draw a ticket without replacement and call it X.
- You draw a second ticket and call it Y.
- What is the covariance of X and Y?

Table for X and Y

x	y	$P(X = x \text{ and } Y = y)$
0	4	$1/4 * 2/3 = 1/6$
0	8	$1/4 * 1/3 = 1/12$
4	0	$1/2 * 1/3 = 1/6$
4	4	$1/2 * 1/3 = 1/6$
4	8	$1/2 * 1/3 = 1/6$
8	0	$1/4 * 1/3 = 1/12$
8	4	$1/4 * 2/3 = 1/6$

Covariance Example 1 Part 1

- First:
  - What is the expected value of X? (Answer: 4.)
  - What is the expected value of Y? (Answer: 4.)
- Second:
  - What are the possible values and probabilities of $(X-4)(Y-4)$?
- Third:
  - What is $\mathrm{Cov}(X, Y)$?

Covariance Example 1 Part 2

Possible Value	Probability
0	5/6
-16	1/6

- Hence $\mathrm{Cov}(X, Y) = 0 \times (5/6) + (-16) \times (1/6) = -8/3$

Covariance Example 2

- Suppose a box has tickets $0, 4, 4, 8$.
- You draw a ticket with replacement and call it X.
- You then draw a second ticket and call it Y.
- What is the covariance of X and Y?

Covariance Example 2 Part 1

- First:
  - What is the expected value of X? (Answer: 4)
  - What is the expected value of Y? (Answer: 4)
- Second:
  - What are the possible values of $(X-4)(Y-4)$?
  - What are the probabilities?
- Third:
  - What is $\mathrm{Cov}(X, Y)$?

Covariance Example 2 Part 2

Possible Value	Probability
-16	1/8
0	3/4
16	1/8

- Hence

$$\mathrm{Cov}(X, Y) = -16 * (1/8) + 0 * (3/4) + 16 * (1/8) = 0.$$

- Is this surprising?

Useful Formulas For Covariance

- These formulas are analogous for the formulas for covariances and variances with lists of numbers.
- First of all, we have the following shortcut formulas:
  - $\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y)$
  - $\mathrm{Var}(X) = \mathrm{Cov}(X, X) = E(X^2) - (E(X))^2$
- If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$.
  - If $\mathrm{Cov}(X, Y) = 0$, however, then X and Y are not necessarily independent.

More Useful Formulas for Variance

- These are again analogous to formulas for the variance of a list of numbers.
  - $\mathrm{Var}(c + X) = \mathrm{Var}(X)$
  - $\mathrm{Var}(cX) = c^2\mathrm{Var}(X)$
  - $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$
- If X and Y are independent, then

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

- The formula above is very important.

Useful Formulas for the Standard Error

- These are analogous to formulas for the standard deviation of a list of numbers.
- They follow from taking the square root of the formulas for the variance.
  - $SE(c + X) = SE(X)$
  - $SE(cX) = |c|SE(X)$
  - If X and Y are independent, then $SE(X + Y) = \sqrt{[SE(X)]^2 + [SE(Y)]^2}$.

Interpretation of Covariance

- $\mathrm{Cov}(X, Y)$ can be thought of as the long-run covariance of X and Y.
  - We imagine creating a very large number of (X, Y) pairs.
- $\mathrm{Corr}(X, Y)$ can be thought of as the long-run correlation of X and Y.
- If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$.
- If $\mathrm{Cov}(X, Y) = 0$, then X and Y may **or may not** be independent.

Covariance Using R

```
#Samples Y for a given value of X
sample_y_x = function(x) {
  p = runif(1)
  if (x==0) {return(4*(p<2/3)+8*(p>=2/3))}
  else if (x==4)
  {return(0*(p<1/3)+4*(p>=1/3 & p<2/3)+8*(p>=2/3))}
  else {return(0*(p<1/3)+4*(p>=1/3)) }
}
```

Covariance Using R Part 2

```
#Samples X
p = runif(1000000)
X = 0*(p<1/4) + 4*(p>=1/4 & p<3/4) + 8*(p>=3/4)
#Samples Y for each value of X
Y=sapply(X,sample_y_x)
#Cov(X,Y)
print(-8/3)
```

```
## [1] -2.666667
```

```
#Should be approximately Cov(X,Y)
print(cov(X,Y))
```

```
## [1] -2.661923
```

Covariance using R Part 3

```r
#Y independent of X
p = runif(1000000)
Y = 0*(p<1/4) + 4*(p>=1/4 & p<3/4) + 8*(p>=3/4)
#Should be approximately 0
print(cov(X,Y))
```

```
## [1] -0.002063606
```

Discussion Question 2

- Suppose a box contains 4 tickets, each with two numbers on it, one on the left, and one on the right.
- The four pairs of numbers are: $(0,0), (0,1), (1,0), (1,1)$.
- You randomly draw one ticket.
  - Let X be the number on the right, and Y the number on the left.
- Are X and Y independent?
- What is $\mathrm{Cov}(X, Y)$?

Discussion Question 3

- Suppose a box contains 4 tickets, each with two numbers on it, one on the left, and one on the right.
- The four pairs of numbers are: $(0, 0), (1, 1), (2, 1), (3, 0)$.
- You randomly draw one ticket.
  - Let X be the number on the right, and Y the number on the left.
- Are X and Y independent?
- What is $\mathrm{Cov}(X, Y)$?

Review Questions

- See Review_questions13.pdf on Canvas.

Reading Summary - Chapter 16

Section 16.1: What Does the Law of Averages Say?

- If you flip a coin n times, the number of heads will be about $n/2$.
- The **absolute difference** between the actual number of heads and $n/2$ tends to get bigger with more flips.
- The **percentage difference** between the actual percentage of heads and 50% tends to get smaller with more flips.

Section 16.2: Chance Processes

- Statisticians are often interested in chance variability.
- To learn about chance variability in a chance process, first make a box model.

Section 16.3: The Sum of Draws

- Many chance processes are analogous to drawing tickets from a box, and taking the sum.
- Sometimes the draws are with replacement, sometimes without.

Section 16.4: Making a Box Model

- Detailed example of making a box model for roulette.

Flipping a Coin

- Suppose you flip a coin n times and it comes up heads k times.
  - The **expected number** of heads is $n/2$.
  - The **expected percentage** of heads is 50%.
  - The **observed number** of heads is k.
  - The **observed percentage** of heads is $(k/n) \times 100\%$.
  - The **absolute chance error** is $|k - n/2|$.
  - The **absolute percentage chance error** is $|(k/n) \times 100\% - 50\%|$.

Law of Averages

- The law of averages says. . .
  - The absolute chance error tends to get larger the more times you flip the coin.
  - The absolute percentage chance error tends to get smaller the more times you flip the coin.

The Law of Averages Using R

```r
#Simulates n fair coin tosses.
#1=heads, 0=tails
cointoss = function(n){
  #n random numbers between 0 and 1
  p = runif(n)
  #Heads if p<=0.5, tails otherwise
  return(1*(p<=0.5))
}
```

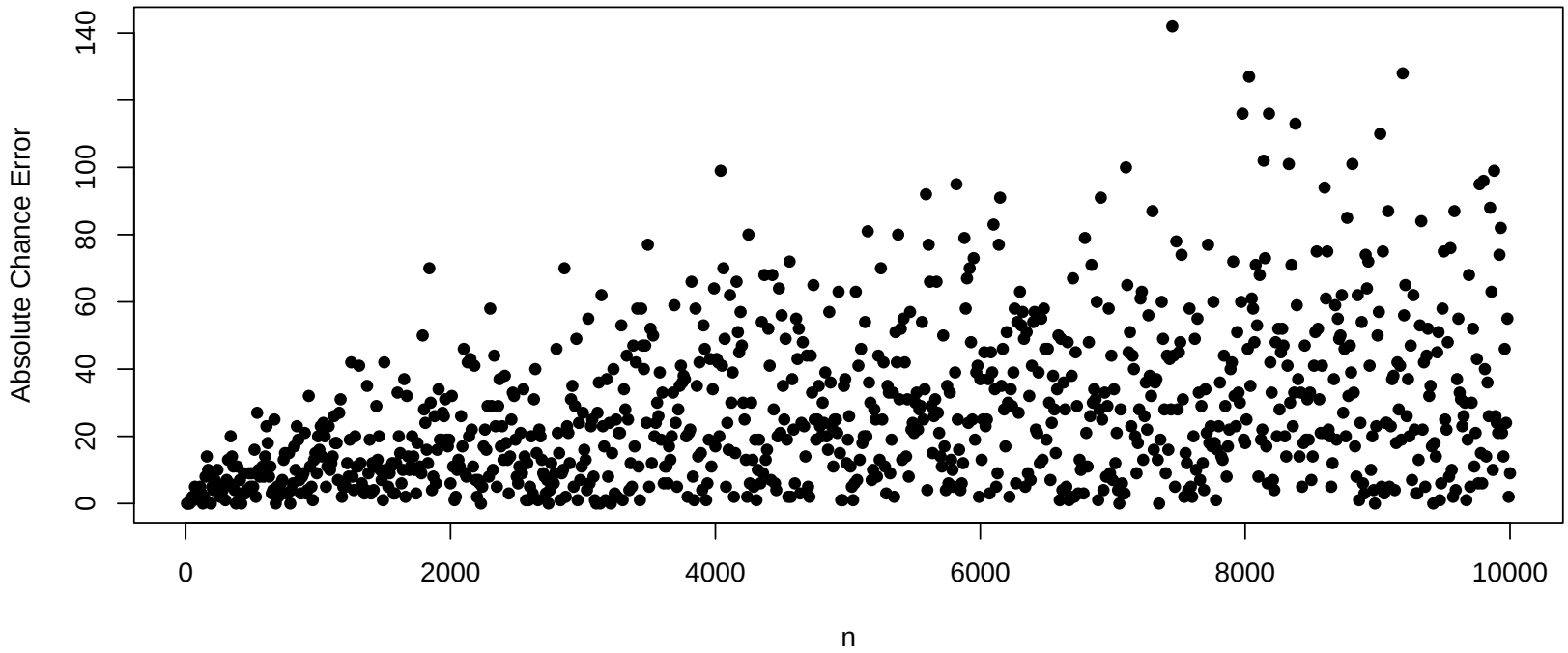
The Law of Averages Using R Part 2

```
#Simulates n fair coin tosses for
#n between 10 and 1,000 and calculates
#the absolute and percent chance error

#Sequence from 10 to 10,000 by 10
numtosses = seq(10,10000,10)
abs_error = c() #Empty vector
pct_error = c() #Empty vector
for (n in numtosses) {
  #We can get the number of heads
  #by summing the tosses
  num_heads = sum(cointoss(n))
  #Appends new absolute error to the end
  abs_error=c(abs_error,abs(num_heads-n/2))
  #Appends new percentage error to the end
  pct_error=c(pct_error,abs((num_heads/n)*100-50))
}
```

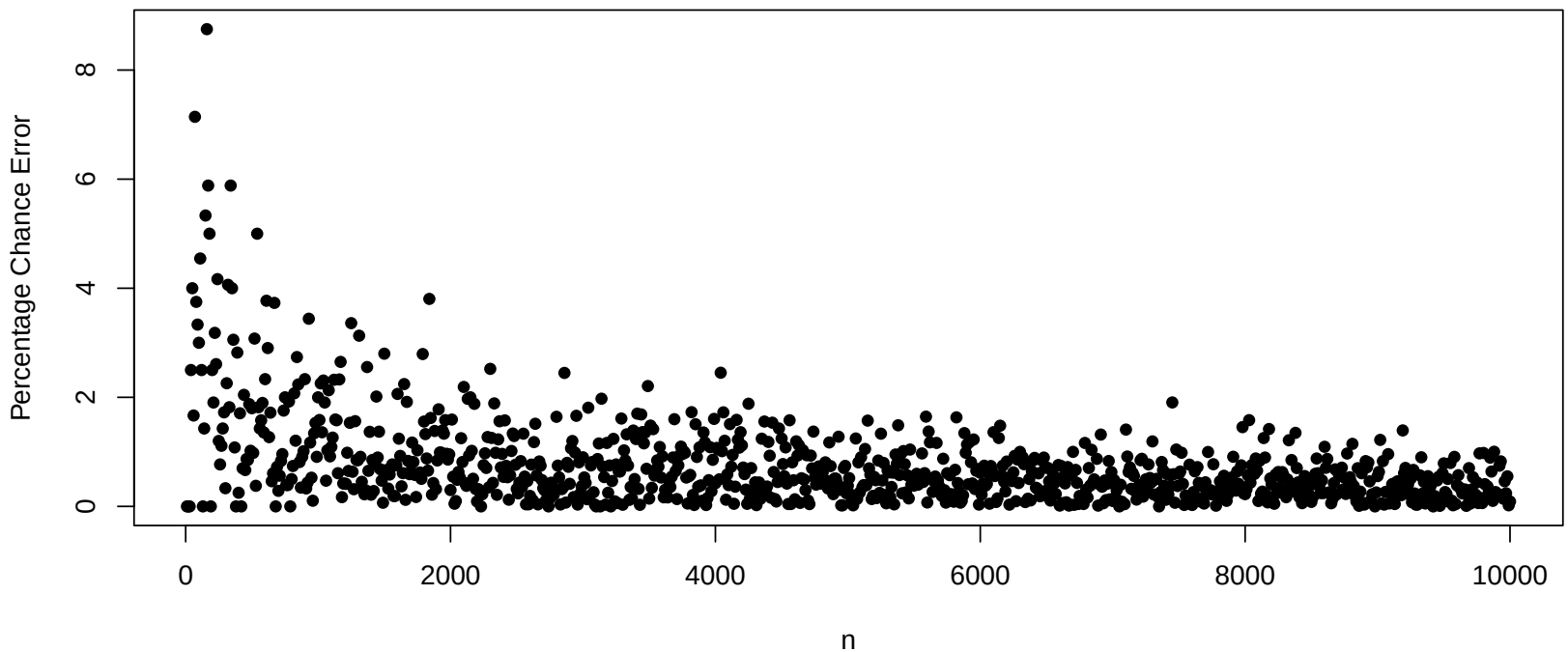
The Law of Averages Using R Part 3

```
#Plots absolute chance error
plot(numtosses,abs_error,xlab='n',
     ylab='Absolute Chance Error',
     pch=16)
```



The Law of Averages Using R Part 4

```
#Plots percentage chance error
plot(numtosses,pct_error,xlab='n',
     ylab='Percentage Chance Error',
     pch=16)
```



Review Questions

- No review questions for this lecture
- Next review questions for Lecture 15